THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 7: **Work, Energy and Simple Machines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A load of 150 N is raised by an effort of 50 N by a machine. The MA is: 1

(a) 200 **(b)** 100 **(c)** 3 **(d)** 1/3

2. A pendulum bob of mass 0.1 kg is raised 0.2 m above its lowest point. Its speed at the lowest point is: 1

(a) 2 m/s **(b)** 0.2 m/s **(c)** 4 m/s **(d)** 10 m/s

3. If the velocity of a body is doubled, its kinetic energy becomes: 1

(a) same **(b)** two times **(c)** four times **(d)** half

4. A man does 4000 J of work in 20 s. The power is: 1

(a) 4020 W **(b)** 200 W **(c)** 80000 W **(d)** 0.005 W

5. A ramp 4 m long raises a load to 1 m height. The ideal MA is: 1

(a) 4 **(b)** 3 **(c)** 5 **(d)** 1/4

6. A body thrown upward with 30 m/s reaches a maximum height of: 1

(a) 90 m **(b)** 45 m **(c)** 30 m **(d)** 15 m

7. If the mass is doubled and the velocity is halved, the KE becomes: 1

(a) one-fourth **(b)** half **(c)** same **(d)** double

8. A 60 W bulb glows for 5 hours. The energy used, in kWh, is: 1

(a) 12 **(b)** 0.12 **(c)** 0.3 **(d)** 3

9. A machine does 400 J of useful work for an energy input of 500 J. Its efficiency is: 1

(a) 125 per cent **(b)** 100 per cent **(c)** 80 per cent **(d)** 20 per cent

10. A spring stores 2 J when compressed by 10 cm. This is: 1

(a) heat **(b)** kinetic energy

(c) sound **(d)** elastic potential energy

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does a heavier truck take a longer distance to stop than a car moving at the same speed with the same braking force? 2

12. Why is the efficiency of a machine always less than 100 per cent? 2

13. Find the speed of a 0.2 kg stone dropped from 5 m just before it reaches the ground (g = 10). 2

14. State why the work done in lifting a body is the same whether it is lifted vertically or pulled up a smooth ramp. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The bob of a pendulum is released from a height 0.45 m above the lowest point. Find the speed at the lowest point (g = 10) and state the energy conversion. 3

16. A body of mass 3 kg is dropped from 20 m. Find the time and KE after 1 s (g = 10). 3

17. Why does a ball bounce lower after each bounce? Use energy ideas. 3

OR

A cyclist and his bicycle of total mass 80 kg move at 5 m/s and then at 10 m/s. Find the work done in increasing the speed. 3

18. State two examples to show that work done can be negative and zero. Calculate the work done by the Earth on the Moon in a circular orbit. 3

OR

A force of 6 N acts on a body as it moves 2 m, then 4 N for 3 m in the same direction. Find the total work done. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A man pulls a trolley of mass 50 kg up a ramp of length 10 m and height 2 m with a force of 150 N. Find the work done by the man, the work done against gravity, the work against friction and the efficiency (g = 10). 5

OR

Explain why the speed of a body sliding down a smooth slide depends only on the height and not on the shape or the mass, with the speeds for h = 5 m and h = 20 m. 5

20. Derive the mechanical advantage of a lever from the principle of moments and calculate the effort for a 600 N load at 20 cm using a 1.5 m bar with the fulcrum at 20 cm from the load. 5

OR

A bullet of mass 20 g fired at 250 m/s into a block of wood stops after 10 cm. Find the average force and the time of penetration. Compare with a repeat at 500 m/s. 5

END OF QUESTION PAPER — Total: 40 Marks

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